

HAMADA, Japan

WMO: 477550

Lat: 34.8967N

Lon: 132.0706E

Elev: 21

StdP: 101.08

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.1	(c) 1.0	(d) -7.5	(e) 2.0	(f) 3.2	(g) -6.3	(h) 2.2	(i) 3.7	(j) 14.1	(k) 7.4	(l) 12.3	(m) 5.7	(n) 4.9	(o) 60	(p) 0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.3	(c) 32.1	(d) 25.7	(e) 31.0	(f) 25.6	(g) 30.1	(h) 25.3	(i) 26.8	(j) 30.2	(k) 26.2	(l) 29.6	(m) 25.7	(n) 29.1	(o) 3.8	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 25.9	(b) 21.3	(c) 29.0	(d) 25.2	(e) 20.4	(f) 28.5	(g) 24.7	(h) 19.8	(i) 28.1	(j) 84.5	(k) 30.3	(l) 81.9	(m) 29.7	(n) 79.6	(o) 29.2	(p) 29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.2	(b) 9.5	(c) 8.3	(e) -2.0	(f) 34.7	(g) 1.6	(h) 1.5	(i) -3.2	(j) 35.7	(k) -4.1	(l) 36.6	(m) -5.0	(n) 37.4	(o) -6.1	(p) 38.5		
(a) 11.2	(b) 9.5	(c) 8.3	(e) -3.6	(f) 27.7	(g) 1.4	(h) 0.7	(i) -4.6	(j) 28.2	(k) -5.4	(l) 28.6	(m) -6.2	(n) 29.0	(o) -7.2	(p) 29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	6.0	6.4	8.9	13.4	17.9	21.4	25.8	27.1	23.0	18.0	13.2	8.5
	DBStd	7.63	2.42	2.78	3.08	3.11	2.44	2.22	2.27	1.94	2.57	2.52	2.76	2.76
	HDD10.0	364	126	108	59	6	0	0	0	0	0	0	4	62
	HDD18.3	1698	382	336	293	151	38	2	0	0	1	37	156	304
	CDD10.0	2497	2	6	24	109	243	343	489	529	389	247	99	16
	CDD18.3	789	0	0	0	3	23	95	231	271	140	25	1	0
	CDH23.3	6244	0	0	1	11	77	342	2070	2824	847	71	1	0
	CDH26.7	1841	0	0	0	1	6	43	610	1005	169	8	0	0
Wind	WSAvg	3.7	4.6	4.2	4.2	3.8	3.3	2.9	3.3	3.0	3.1	3.4	3.9	4.5
Precipitation	PrecAvg	1649	102	78	123	108	161	175	253	132	183	107	116	99
	PrecMax	2375	153	138	197	219	283	393	638	434	330	308	184	197
	PrecMin	1127	33	39	56	32	78	19	9	24	83	23	34	20
	PrecStd	332	34	30	46	54	57	108	176	99	83	64	47	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	18.6	20.9	24.9	27.4	29.5	33.0	34.6	31.6	27.6	22.8	17.5
		MCWB	12.4	13.4	14.1	17.3	19.0	23.5	25.5	25.2	25.2	22.4	17.4	13.4
	2%	DB	13.0	15.3	18.1	22.2	25.3	28.0	31.5	32.8	29.8	25.1	20.8	15.7
		MCWB	9.3	11.7	12.4	15.7	18.4	22.9	25.7	25.6	24.8	20.4	16.4	11.7
	5%	DB	11.6	13.0	16.1	20.4	23.8	26.5	30.5	31.6	28.6	23.7	19.5	14.4
		MCWB	7.9	9.5	11.7	14.7	17.5	21.8	25.3	25.8	24.1	19.2	15.3	10.5
	10%	DB	10.3	11.2	14.5	18.9	22.6	25.2	29.6	30.7	27.4	22.5	18.2	13.1
		MCWB	6.8	7.8	10.8	14.0	17.0	21.2	25.0	25.7	23.4	18.3	14.3	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	14.5	15.6	19.4	21.6	24.7	27.5	27.7	26.3	23.2	18.9	14.9
		MCDB	14.8	17.2	18.4	22.8	24.6	27.6	30.6	30.9	29.4	26.3	21.2	16.8
	2%	WB	10.3	12.3	13.9	17.2	20.1	23.8	26.6	27.1	25.5	21.8	17.5	12.7
		MCDB	12.4	14.8	16.6	20.3	23.0	26.5	29.8	30.5	28.6	24.3	19.7	14.8
	5%	WB	8.3	10.1	12.6	15.8	19.2	23.1	26.0	26.6	24.7	20.4	16.2	11.2
		MCDB	10.8	12.3	15.3	18.8	22.1	25.4	29.3	30.1	27.8	22.7	18.7	13.6
	10%	WB	7.1	8.4	11.3	14.9	18.3	22.3	25.5	26.1	24.0	19.1	14.9	9.9
		MCDB	9.8	10.6	14.0	17.8	21.2	24.3	28.7	29.6	26.6	21.5	17.6	12.5
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.7	7.8	8.3	7.9	6.4	5.6	6.3	6.7	7.6	7.2	6.0
		MCDBR	8.2	9.8	10.2	10.3	9.9	7.8	7.0	7.7	7.3	8.7	8.8	8.2
		MCWBR	6.1	7.4	7.2	6.5	5.8	4.2	3.5	3.7	4.0	5.1	5.9	5.9
	5% WB	MCDBR	7.6	8.8	9.1	8.4	7.8	5.8	6.3	6.5	6.5	7.2	7.7	7.4
		MCWBR	6.3	7.2	7.2	5.9	5.3	3.3	3.6	3.7	3.9	4.6	5.8	6.1
Clear-Sky Solar Irradiance	taub	0.439	0.481	0.552	0.576	0.572	0.591	0.588	0.547	0.503	0.460	0.440	0.430	
	taud	2.006	1.932	1.785	1.781	1.843	1.879	1.939	2.058	2.121	2.128	2.090	2.043	
	Ebn at Noon	723	740	723	733	742	727	725	748	760	756	720	703	
	Edh at Noon	137	165	208	218	208	200	187	164	147	134	125	125	
All-Sky Solar Radiation	RadAvg	1.97	2.71	3.82	4.82	5.36	4.84	4.94	5.14	4.05	3.38	2.40	1.79	
	RadStd	0.15	0.25	0.31	0.40	0.56	0.43	0.71	0.53	0.37	0.32	0.20	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page